

```
import { useState, useRef, useEffect } from "react";
```

```
const LYRIC_SPEC = `You are LYRIC_ENGINE v1.0. Generate lyrics based on this spec
```

```
LYRIC_MODE_DECISION:
```

- Infer genre and topic from user_request
- If user explicitly requests "poetic" or "repetitive", use that mode
- Otherwise use this mapping:
 - * Genre in {pop, mainstream pop, dance-pop, hyperpop, hip-hop, rap, trap, R&B,
 - * Genre in {rock, alt rock, indie, indie rock, folk, soul, jazz, blues, metal,
 - * Ambiguous: if user emphasizes "catchy", "radio", "hit", "club", "anthem" -> R

```
POETIC MODE:
```

- Aim for score $\geq 9/10$ from a ruthless critic
- High originality, minimal clichés
- Strong, evocative imagery and creative metaphors
- Clear theme or narrative
- At least 2 standalone quotable lines
- Flexible structure: [Verse], [Chorus], [Bridge] or free verse
- Match tone to genre: folk/singer-songwriter = reflective, subtle; rock/metal =

```
REPETITIVE MODE:
```

- Optimized for chart potential
- Use section headers: [Verse 1], [Pre-Chorus], [Chorus], [Verse 2], [Bridge]
- Core hook line ≤ 10 words, repeat 3-8 times total
- 30-55% of lyrics = repeated lines
- Use title phrase verbatim in chorus ≥ 2 times
- Short punchy lines, Grade 3-7 readability
- First and second person: "I", "you", "we"
- Concrete imagery, one main idea
- ≥ 2 quotable lines (TikTok caption worthy)
- Under 250 words total

```
SELF-EVAL LOOP:
```

- After draft, internally simulate critic pass
- If score $< 8/10$ or obvious clichés or < 2 quotable lines: rewrite
- Repeat until score $\geq 8/10$ and no rule violations

```
OUTPUT FORMAT:
```

- Return ONLY the lyrics text
- No commentary, no explanations, no scores, no mode disclosure
- Only include section headers if part of lyrics structure`;

```
const GENRES = [
```

```

    { label: "Pop", value: "pop", emoji: "🌟" },
    { label: "Hip-Hop", value: "hip-hop", emoji: "🎤" },
    { label: "Indie", value: "indie", emoji: "🎸" },
    { label: "Folk", value: "folk", emoji: "🎵" },
    { label: "R&B", value: "R&B", emoji: "🎶" },
    { label: "Rock", value: "rock", emoji: "🎸" },
    { label: "K-Pop", value: "K-pop", emoji: "🎤" },
    { label: "Jazz", value: "jazz", emoji: "🎷" },
  ];

const MOODS = [
  { label: "Heartbreak", value: "heartbreak", color: "#e55" },
  { label: "Euphoric", value: "euphoric", color: "#f90" },
  { label: "Nostalgic", value: "nostalgic", color: "#a8d" },
  { label: "Defiant", value: "defiant", color: "#f55" },
  { label: "Longing", value: "longing", color: "#58f" },
  { label: "Hopeful", value: "hopeful", color: "#4c8" },
  { label: "Melancholic", value: "melancholic", color: "#78a" },
  { label: "Sensual", value: "sensual", color: "#e8a" },
];

export default function LyricEngine() {
  const [topic, setTopic] = useState("");
  const [genre, setGenre] = useState("");
  const [mood, setMood] = useState("");
  const [lyrics, setLyrics] = useState("");
  const [loading, setLoading] = useState(false);
  const [error, setError] = useState("");
  const [copied, setCopied] = useState(false);
  const [charCount, setCharCount] = useState(0);
  const lyricsRef = useRef(null);
  const textareaRef = useRef(null);

  useEffect(() => {
    setCharCount(topic.length);
  }, [topic]);

  useEffect(() => {
    if (lyrics && lyricsRef.current) {
      lyricsRef.current.scrollToView({ behavior: "smooth", block: "start" });
    }
  }, [lyrics]);

  const buildPrompt = () => {
    let prompt = topic;
    if (genre) prompt += ` . Genre: ${genre}`;
    if (mood) prompt += ` . Mood/tone: ${mood}`;
  };

```

```

    return prompt;
};

const generate = async () => {
  if (!topic.trim()) return;
  setLoading(true);
  setError("");
  setLyrics("");

  try {
    const response = await fetch("https://api.anthropic.com/v1/messages", {
      method: "POST",
      headers: {
        "Content-Type": "application/json",
        "anthropic-dangerous-direct-browser-ipc": "true",
      },
      body: JSON.stringify({
        model: "claude-sonnet-4-20250514",
        max_tokens: 1000,
        system: LYRIC_SPEC,
        messages: [{ role: "user", content: buildPrompt() }],
      }),
    });

    const data = await response.json();
    const text = data.content?.map((b) => b.text || "").join("") || "";
    if (!text) throw new Error("No lyrics returned");
    setLyrics(text.trim());
  } catch (e) {
    setError("Something went wrong. Please try again.");
  } finally {
    setLoading(false);
  }
};

const handleCopy = () => {
  navigator.clipboard.writeText(lyrics);
  setCopied(true);
  setTimeout(() => setCopied(false), 2000);
};

const handleKeyDown = (e) => {
  if (e.key === "Enter" && (e.metaKey || e.ctrlKey)) generate();
};

const selectedMoodColor = MOODS.find((m) => m.value === mood)?.color || "#c9a96"

```

```

return (
  <div style={styles.root}>
    {/* Grain overlay */}
    <div style={styles.grain} />

    {/* Header */}
    <header style={styles.header}>
      <div style={styles.logoRow}>
        <span style={styles.logoSymbol}>J</span>
        <span style={styles.logoText}>LYRIC ENGINE</span>
        <span style={styles.logoVersion}>v1.0</span>
      </div>
      <p style={styles.tagline}>สร้างเนื้อเพลงด้วย AI · Write songs that matter</p>
    </header>

    <main style={styles.main}>
      {/* Input section */}
      <section style={styles.card}>
        <label style={styles.label}>
          เกี่ยวกับเพลงของคุณ <span style={styles.labelSub}>)/ What's your song about
        </label>
        <div style={styles.textareaWrapper}>
          <textarea
            ref={textareaRef}
            style={styles.textarea}
            placeholder="เช่น: เพลงเกี่ยวกับการเดินทางคนเดียวในคืนฝนตก และความรู้สึกที่ไม่แน่ใจว่า"
            value={topic}
            onChange={(e) => setTopic(e.target.value)}
            onKeyDown={handleKeyDown}
            maxLength={300}
            rows={4}
          />
          <span style={styles.charCount}>{charCount}/300</span>
        </div>

        {/* Genre picker */}
        <div style={styles.sectionLabel}>แนวเพลง · Genre</div>
        <div style={styles.chipRow}>
          {GENRES.map((g) => (
            <button
              key={g.value}
              style={{
                ...styles.chip,
                ...(genre === g.value ? styles.chipActive : {}),
              }}
              onClick={() => setGenre(genre === g.value ? "" : g.value)}
            >

```

```

        {g.emoji} {g.label}
      </button>
    )))
  </div>

  {/* Mood picker */}
  <div style={styles.sectionLabel}>อารมณ์ · Mood</div>
  <div style={styles.chipRow}>
    {MOODS.map((m) => (
      <button
        key={m.value}
        style={{
          ...styles.chip,
          ...(mood === m.value
            ? { ...styles.chipActive, borderColor: m.color, color: m.color
              : {}),
          }}
        onClick={() => setMood(mood === m.value ? "" : m.value)}
      >
        {m.label}
      </button>
    )))
  </div>

  <button
    style={{
      ...styles.generateBtn,
      ...(loading || !topic.trim() ? styles.generateBtnDisabled : {}),
    }}
    onClick={generate}
    disabled={loading || !topic.trim()}
  >
    {loading ? (
      <span style={styles.loadingRow}>
        <span style={styles.spinner} />
        กำลังแต่งเพลง...
      </span>
    ) : (
      "+ สร้างเนื้อเพลง · Generate"
    )}
  </button>

  <p style={styles.hint}>* + Enter เพื่อสร้างตัววน</p>
</section>

{/* Error */}
{error && <div style={styles.error}>{error}</div>}
```

```

    { /* Lyrics output */ }
    { lyrics && (
      <section ref={lyricsRef} style={styles.lyricsCard}>
        <div style={styles.lyricsHeader}>
          <span style={styles.lyricsTitle}>> เนื้อเพลงของคุณ</span>
          <button style={styles.copyBtn} onClick={handleCopy}>
            {copied ? "✓ คัดลอกแล้ว" : "คัดลอก"}
          </button>
        </div>
        <div style={styles.lyricsBody}>
          {lyrics.split("\n").map((line, i) => {
            const isHeader = line.match(/^\[.\+\]\$/);
            if (!line.trim()) return <div key={i} style={styles.lyricBlank} /
            return (
              <div
                key={i}
                style={isHeader ? styles.lyricHeader : styles.lyricLine}
              >
                {line}
              </div>
            );
          })}
        </div>
        <button
          style={styles.regenerateBtn}
          onClick={generate}
          disabled={loading}
        >
          🔄 สร้างใหม่ · Regenerate
        </button>
      </section>
    )}
  </main>

  <footer style={styles.footer}>
    Powered by Claude · LYRIC_ENGINE SPEC v1.0
  </footer>
</div>
);
}

const styles = {
  root: {
    minHeight: "100vh",
    background: "#0e0c0a",
    color: "#e8dcc8",
  },

```

```
    fontFamily: "'Georgia', 'Times New Roman', serif",
    position: "relative",
    overflowX: "hidden",
  },
  grain: {
    position: "fixed",
    inset: 0,
    backgroundImage: `url("data:image/svg+xml,%3Csvg viewBox='0 0 256 256' xmlns=
    pointerEvents: "none",
    zIndex: 0,
  },
  header: {
    textAlign: "center",
    padding: "60px 24px 32px",
    position: "relative",
    zIndex: 1,
    borderBottom: "1px solid #2a2520",
  },
  logoRow: {
    display: "flex",
    alignItems: "baseline",
    justifyContent: "center",
    gap: "12px",
    marginBottom: "8px",
  },
  logoSymbol: {
    fontSize: "32px",
    color: "#c9a96e",
    fontStyle: "normal",
  },
  logoText: {
    fontSize: "28px",
    fontWeight: "700",
    letterSpacing: "0.3em",
    color: "#e8dcc8",
    fontFamily: "'Georgia', serif",
  },
  logoVersion: {
    fontSize: "11px",
    color: "#6a5f4e",
    letterSpacing: "0.2em",
    fontFamily: "monospace",
  },
  tagline: {
    fontSize: "13px",
    color: "#7a6f5e",
    letterSpacing: "0.1em",
```

```
    margin: 0,
    fontStyle: "italic",
  },
  main: {
    maxWidth: "680px",
    margin: "0 auto",
    padding: "40px 24px 60px",
    position: "relative",
    zIndex: 1,
    display: "flex",
    flexDirection: "column",
    gap: "24px",
  },
  card: {
    background: "#161310",
    border: "1px solid #2a2520",
    borderRadius: "4px",
    padding: "32px",
  },
  label: {
    display: "block",
    fontSize: "14px",
    fontWeight: "600",
    color: "#c9a96e",
    letterSpacing: "0.05em",
    marginBottom: "12px",
    fontFamily: "monospace",
  },
  labelSub: {
    color: "#6a5f4e",
    fontWeight: "400",
    fontStyle: "italic",
    fontFamily: "'Georgia', serif",
  },
  textareaWrapper: {
    position: "relative",
    marginBottom: "28px",
  },
  textarea: {
    width: "100%",
    background: "#0e0c0a",
    border: "1px solid #2a2520",
    borderRadius: "2px",
    color: "#e8dcc8",
    fontSize: "15px",
    fontFamily: "'Georgia', serif",
    lineHeight: "1.7",
```



```
padding: "16px",
resize: "vertical",
outline: "none",
boxSizing: "border-box",
transition: "border-color 0.2s",
},
charCount: {
  position: "absolute",
  bottom: "8px",
  right: "12px",
  fontSize: "11px",
  color: "#4a4035",
  fontFamily: "monospace",
},
sectionLabel: {
  fontSize: "11px",
  color: "#6a5f4e",
  letterSpacing: "0.15em",
  fontFamily: "monospace",
  marginBottom: "10px",
  textTransform: "uppercase",
},
chipRow: {
  display: "flex",
  flexWrap: "wrap",
  gap: "8px",
  marginBottom: "20px",
},
chip: {
  background: "transparent",
  border: "1px solid #2a2520",
  borderRadius: "2px",
  color: "#7a6f5e",
  fontSize: "12px",
  padding: "6px 12px",
  cursor: "pointer",
  fontFamily: "monospace",
  letterSpacing: "0.05em",
  transition: "all 0.15s",
},
chipActive: {
  background: "#c9a96e22",
  borderColor: "#c9a96e",
  color: "#c9a96e",
},
generateBtn: {
  width: "100%",
```

```
    background: "#c9a96e",
    color: "#0e0c0a",
    border: "none",
    borderRadius: "2px",
    fontSize: "14px",
    fontWeight: "700",
    letterSpacing: "0.1em",
    padding: "16px",
    cursor: "pointer",
    fontFamily: "monospace",
    transition: "all 0.2s",
    marginTop: "8px",
  },
  generateBtnDisabled: {
    background: "#2a2520",
    color: "#4a4035",
    cursor: "not-allowed",
  },
  loadingRow: {
    display: "flex",
    alignItems: "center",
    justifyContent: "center",
    gap: "10px",
  },
  spinner: {
    width: "14px",
    height: "14px",
    border: "2px solid #4a4035",
    borderTopColor: "#0e0c0a",
    borderRadius: "50%",
    display: "inline-block",
    animation: "spin 0.8s linear infinite",
  },
  hint: {
    textAlign: "center",
    fontSize: "11px",
    color: "#3a3028",
    fontFamily: "monospace",
    marginTop: "10px",
    marginBottom: 0,
  },
  error: {
    background: "#2a1010",
    border: "1px solid #5a2020",
    borderRadius: "2px",
    color: "#e87070",
    padding: "16px",
```

```
    fontSize: "13px",
    fontFamily: "monospace",
  },
  lyricsCard: {
    background: "#0e0c0a",
    border: "1px solid #c9a96e44",
    borderRadius: "4px",
    padding: "32px",
    position: "relative",
  },
  lyricsHeader: {
    display: "flex",
    justifyContent: "space-between",
    alignItems: "center",
    marginBottom: "28px",
    paddingBottom: "16px",
    borderBottom: "1px solid #2a2520",
  },
  lyricsTitle: {
    fontSize: "13px",
    color: "#c9a96e",
    fontFamily: "monospace",
    letterSpacing: "0.1em",
  },
  copyBtn: {
    background: "transparent",
    border: "1px solid #2a2520",
    color: "#7a6f5e",
    fontSize: "11px",
    padding: "5px 12px",
    borderRadius: "2px",
    cursor: "pointer",
    fontFamily: "monospace",
    letterSpacing: "0.05em",
    transition: "all 0.15s",
  },
  lyricsBody: {
    lineHeight: "1.9",
    marginBottom: "28px",
  },
  lyricLine: {
    fontSize: "16px",
    color: "#e8dcc8",
    fontFamily: "'Georgia', serif",
    fontStyle: "italic",
  },
  lyricHeader: {
```

```

        fontSize: "11px",
        color: "#c9a96e",
        fontFamily: "monospace",
        letterSpacing: "0.2em",
        marginTop: "20px",
        marginBottom: "4px",
        fontStyle: "normal",
    },
    lyricBlank: {
        height: "8px",
    },
    regenerateBtn: {
        background: "transparent",
        border: "1px solid #2a2520",
        color: "#6a5f4e",
        fontSize: "12px",
        padding: "10px 20px",
        borderRadius: "2px",
        cursor: "pointer",
        fontFamily: "monospace",
        letterSpacing: "0.1em",
        transition: "all 0.2s",
        width: "100%",
    },
    footer: {
        textAlign: "center",
        padding: "24px",
        fontSize: "10px",
        color: "#3a3028",
        fontFamily: "monospace",
        letterSpacing: "0.15em",
        borderTop: "1px solid #1a1710",
        position: "relative",
        zIndex: 1,
    },
},
};

// Inject spinner keyframe
const styleTag = document.createElement("style");
styleTag.textContent = `
    @keyframes spin { to { transform: rotate(360deg); } }
    textarea:focus { border-color: #c9a96e66 !important; }
    textarea { transition: border-color 0.2s; }
`;
document.head.appendChild(styleTag);

```

